

CISTERN OVERFLOW Alert System

T.K. HAREENDRAN

The system presented here will electronically monitor cistern (tank that holds water to flush the toilet) overflows that can otherwise cause seepage in the flats below in a multi-story building. In case the cistern's mechanism falters and water starts overflowing, it will sound an alarm to alert you with noticeable audible and visual warnings.

The sensor of the system comprises stainless-steel electrodes (needles)

housed in a 2.54cm (1-inch) diameter connector (with a small outlet tube attached) for fitting directly into the side overflow pipe of a concealed flushing cistern that also has an internal overflow arrangement. It can be inserted easily into the overflow hole, and detection level can be set as per requirement. Signals from the sensor can be channelled to the master (alarm) unit with a low-voltage twin cable.

The master unit is housed in a

wall-mounted plastic enclosure. Power supply is provided from a standard 9V battery. Besides, a volt-free switch terminal is provided for optional remote overflow indication to the building's caretaker for a central common alarm.

Fig. 1(a) shows the sensor of the cistern overflow alert system, Fig. 1(b) the sensor made by the author with the help of a 2.54cm diameter bottle cap, two sewing machine needles and one small PVC tube, and Fig 1(c) the shell of a discarded restroom air-freshener used as prototype enclosure, wall-mounted in the author's toilet.

Circuit and working

Circuit diagram of the master unit of the cistern overflow alert system is shown in Fig. 2.

The master unit is built around quad-bilateral switch CD4066 (IC1), in which switches IC1A and IC1B simultaneously extend positive supply to the output. These are driven by the water sensor (connected to J1) through IC1A.

Overflow alert visual indicator

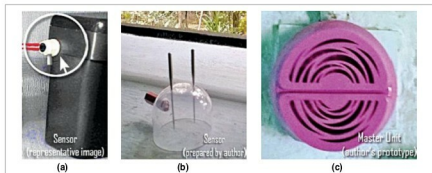


Fig. 1: Author's sensor and prototype: (a) sensor of a cistern overflow alert system, (b) sensor made with the help of a 2.54-cm diameter bottle cap, two sewing machine needles and one small PVC tube, and (c) shell of a discarded restroom air-freshener as prototype enclosure, wall-mounted in a toilet

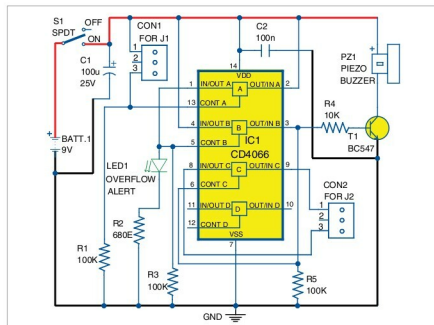


Fig. 2: Circuit diagram of the master unit of the cistern overflow alert system

PARTS LIST

Semiconductors:

- | | |
|------|--------------------------------|
| IC1 | - CD4066 quad-bilateral switch |
| T1 | - BC547 npn transistor |
| LED1 | - 5mm LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|------------|----------------|
| R1, R3, R5 | - 100-kilo-ohm |
| R2 | - 680-ohm |
| R4 | - 10-kilo-ohm |

Capacitors:

- C1 - 100 μ F, 25V electrolytic
C2 - 100nF ceramic disk

Miscellaneous:

- PZ1 - Piezo buzzer

S1

- | | |
|------------|--|
| J1 | - Two copper wires for connecting to two needles (sensors) |
| J2 | - Two copper wires for optional remote indication |
| CON1, CON2 | - 3-pin connector
- 2-pin connector for BATT.1
- 2-pin connector for PZ1
- 9V PP3 battery
- Two sewing machine needles
- One small PVC tube |